Ratsional, irratsional va haqiqiy sonlar

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Ratsional son — bu …**

A) faqat musbat kasr son

B) ildiz belgisi bor son

C) $\frac{a}{b}$ ko'rinishda yozish mumkin bo'lgan son (a, b — butun sonlar, b ≠ 0)

D) faqat butun son

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2. **Qaysi son ratsional emas?**

A) $\frac{5}{8}$ **B)** $\sqrt{3}$ **C)** 0,7 **D)** −12

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3. **−5 sonini $\frac{a}{b}$ ko'rinishida yozing.**

A) $\frac{-5}{0}$ **B)** $\frac{-1}{5}$ **C)** $\frac{-10}{2}$ **D)** $\frac{5}{1}$

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4. **$\frac{3}{20}$ ni o'nli kasrga aylantiring.**

A) 0,(15) **B)** 0,15 **C)** 1,5 **D)** 0,35

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5. **$\frac{8}{9}$ ning o'nli yozuvi qaysi?**

A) 0,8 **B)** 0,(89) **C)** 0,88 **D)** 0,(8)

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6. **Qaysi kasr chekli o'nli kasrga aylanadi?**

A) $\frac{7}{25}$ **B)** $\frac{1}{3}$ **C)** $\frac{5}{6}$ **D)** $\frac{2}{7}$

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7. **0,545454… ni qavs yordamida yozing.**

A) 0,(545) **B)** 0,(54) **C)** 0,5(4) **D)** 0,54

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8. **2,4777… ni qavs yordamida yozing.**

A) 2,47 **B)** 2,(47) **C)** 2,4(7) **D)** 2,(4777)

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9. **0,24 ni qisqarmas oddiy kasr ko'rinishida yozing.**

A) $\frac{12}{50}$ **B)** $\frac{24}{10}$ **C)** $\frac{6}{25}$ **D)** $\frac{1}{24}$

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10. **Qaysi son irratsional?**

A) π **B)** 3,14 **C)** 0,(3) **D)** $\frac{22}{7}$

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11. $\sqrt{64}$ **qanday son?**

A) irratsional
B) haqiqiy, lekin ratsional emas
C) manfiy son
D) ratsional (8 ga teng)

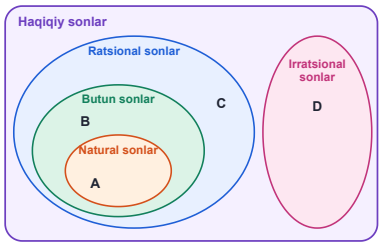
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12. Haqiqiy sonlar nimalardan iborat?

A) faqat kasrlardan
B) faqat butun sonlardan
C) ratsional va irratsional sonlardan
D) faqat musbat sonlardan

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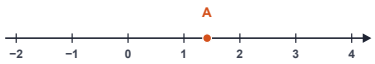
13. Rasmda sonlar to'plamlari, har sohada harf bor. −4 soni qaysi harf turgan sohaga tushadi?



A) C
B) B
C) D
D) A

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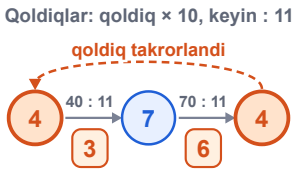
14. Rasmdagi sonlar o'qida A nuqta qaysi songa mos keladi?



A) −1,4
B) π
C) $\sqrt{2}$
D) 2,5

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15. Rasmda 4 ni 11 ga bo'lishdagi qoldiqlar va bo'linma raqamlari. $\frac{4}{11}$ **qanday yoziladi?**



A) 0,36
B) 0,(36)
C) 0,3(6)
D) 0,(4)

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16. Qaysi son butun emas, lekin ratsional?

A) 0
B) −7
C) $\frac{2}{3}$
D) $\sqrt{49}$

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17. Qaysi biri natural son?

A) $\sqrt{25}$
B) 0
C) −2
D) $\frac{1}{5}$

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18. 0,(3) qaysi kasrga teng?

A) $\frac{3}{10}$
B) $\frac{1}{30}$
C) $\frac{1}{3}$
D) $\frac{3}{100}$

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19. 1 ni 8 ga bo'lganda qoldiq 0 bo'lib, bo'lish tugadi. $\frac{1}{8}$ **= ?**

A) 0,(125)
B) 0,125
C) 0,18
D) 0,8

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20. Qaysi tasdiq to'g'ri?

A) Har qanday butun son — ratsional son
B) $\sqrt{2} = 1{,}41$
C) π — ratsional son
D) Har qanday ratsional son — butun son

Javoblar

Ratsional, irratsional va haqiqiy sonlar · 1-test (oson)

№	Javob	Qiymati
1	**C**	$\frac{a}{b}$ ko'rinishda yozish mumkin bo'lgan son (a, b — butun sonlar, b ≠ 0)
2	**B**	$\sqrt{3}$
3	**C**	$\frac{-10}{2}$
4	**B**	0,15
5	**D**	0,(8)
6	**A**	$\frac{7}{25}$
7	**B**	0,(54)
8	**C**	2,4(7)
9	**C**	$\frac{6}{25}$
10	**A**	π

№	Javob	Qiymati
11	**D**	ratsional (8 ga teng)
12	**C**	ratsional va irratsional sonlardan
13	**B**	B
14	**C**	$\sqrt{2}$
15	**B**	0,(36)
16	**C**	$\frac{2}{3}$
17	**A**	$\sqrt{25}$
18	**C**	$\frac{1}{3}$
19	**B**	0,125
20	**A**	Har qanday butun son — ratsional son

Qisqa yechimlar

1. Ta'rif bo'yicha: ikki butun sonning bo'linmasi $\frac{a}{b}$ ko'rinishida yoziladigan son ($b \neq 0$) — **ratsional son**.
Butun sonlar, kasrlar, manfiy sonlar ham shunga kiradi.

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2. −12 = $-\frac{12}{1}$, 0,7 = $\frac{7}{10}$, $\frac{5}{8}$ — hammasi kasr ko'rinishida.
3 to'la kvadrat emas, $\sqrt{3} = 1{,}732\ldots$ — raqamlar takrorlanmaydi. Javob: $\sqrt{3}$.

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3. −10 : 2 = −5. Javob: $\frac{-10}{2}$.
$-\frac{1}{5}$ = −0,2; $\frac{5}{1}$ = 5 (ishora yo'qoldi); maxraj 0 bo'lishi mumkin emas.

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4. 3 : 20 = 0,15 — qoldiq 0 bo'ldi, kasr chekli.
Tekshirish: 0,15 = $\frac{15}{100}$ = $\frac{3}{20}$. Javob: **0,15**.

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5. 8 : 9: har qadamda qoldiq 8, bo'linmada 8 raqami takrorlanaveradi: 0,888…
Qisqa yozuv: **0,(8)**.

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6. 7 : 25 = 0,28 — qoldiq 0, chekli. Javob: $\frac{7}{25}$.
$\frac{1}{3}$ = 0,(3), $\frac{2}{7}$ = 0,(285714), $\frac{5}{6}$ = 0,8(3) — davriy.

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7. Verguldan keyin darhol «54» takrorlanadi.
Takrorlanuvchi raqamlar qavsga olinadi: **0,(54)**. (0,54 — chekli kasr, boshqa son.)

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8. 4 bir marta keladi, keyin faqat 7 takrorlanadi.
Qavsga faqat 7 olinadi: **2,4(7)**.

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9. Ikki xona — maxraj 100: 0,24 = $\frac{24}{100}$.
4 ga qisqartiramiz: $\frac{6}{25}$. ($\frac{12}{50}$ — teng, lekin yana qisqaradi.)

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10. $\frac{22}{7}$ — kasr, 3,14 — chekli, 0,(3) — davriy: hammasi ratsional.

π = 3,14159… — raqamlari takrorlanmaydi, irratsional. $\frac{22}{7}$ va 3,14 — uning taqribiy qiymatlari.

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11. 8 · 8 = 64, demak $\sqrt{64} = 8$ — natural son.

Har qanday natural son ratsional: **ratsional (8 ga teng)**. Ildiz belgisi sonni irratsional qilmaydi.

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12. Ratsional va irratsional sonlar birgalikda **haqiqiy sonlarni** tashkil etadi.

Har bir haqiqiy son — sonlar o'qidagi nuqta.

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13. −4 — butun son, lekin natural emas (natural sonlar 1, 2, 3, …).

U «Butun sonlar» ichida, «Natural sonlar»dan tashqarida: **B**.

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14. A nuqta 1 bilan 2 orasida, 1 ga yaqinroq.

$\sqrt{2} = 1{,}41\ldots$ — mos. $\pi \approx 3{,}14$ va 2,5 — 2 dan o'ngda, −1,4 — 0 dan chapda. Javob: $\sqrt{2}$.

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15. Qoldiqlar 4, 7, 4 — qoldiq takrorlandi, raqamlar 3 va 6 ham qayta-qayta keladi: 0,3636…

Davr verguldan keyin darhol boshlanadi: **0,(36)**.

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16. −7, 0 va $\sqrt{49} = 7$ — butun sonlar.

$\frac{2}{3}$ = 0,(6) — butun emas, lekin kasr ko'rinishida: ratsional.

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17. $\sqrt{25} = 5$, chunki 5 · 5 = 25 — natural son. Javob: $\sqrt{25}$.

−2 va 0 — butun, lekin natural emas; $\frac{1}{5}$ — kasr.

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18. 1 : 3 = 0,333… = 0,(3). Javob: $\frac{1}{3}$.

$\frac{3}{10}$ = 0,3 — chekli kasr, boshqa son.

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19. 1 : 8 = 0,125 — qoldiq 0, kasr chekli. Javob: **0,125**.

0,(125) — davriy, bo'lish tugamaganda yoziladi.

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20. Butun son n = $\frac{n}{1}$ — ratsional: **to'g'ri**.

$\frac{1}{2}$ ratsional, lekin butun emas; π — irratsional; $\sqrt{2} \approx 1{,}41$, aniq teng emas.
